

Final cluster label distribution by count:

0	13740
2	16258
3	16317
4	1511
5	4813
6	678
7	425
business_0	483
business_1	1739
business_2	10
business_3	1294
business_4	414
business_5	153
business_6	3575

1st round CLUSTERS (k=8-1(recluster)=7)

Cluster 0:

High (0.65-1.0): 'multi_country_transactions', 'card_top_mcc_weight',
'median_time_between_transactions', 'std_time_between_transactions',
'transaction_count_EFT', 'Averagetime_between_transaction_day', 'birth_business_age',
'credit_share', 'transaction_count_EMT', 'account_age'
Moderate (0.35-0.65): 'pct_transactions_top10_busiest_days'

Main Province: Ontario

Cluster 1: (Used for Recluster-2nd round cluster)

High (0.65-1.0): 'transaction_count_Cheque', 'debit_transaction_amount',
'credit_transaction_amount', 'average_debit_transaction_amount',
'average_credit_transaction_amount', 'Averagetime_between_transaction_day',
'std_time_between_transactions', 'pct_transactions_top10_busiest_days', 'credit_share',
'median_time_between_transactions'
Moderate (0.35-0.65):

Main Province: Mostly Ontario and Other

Cluster 2:

High (0.65-1.0): 'total_transaction_count', 'transaction_count_ABM',
'transaction_count_Card', 'unique_transaction_cities', 'unique_transaction_provinces',
'multi_country_transactions', 'card_ecommerce_count', 'card_non_ecommerce_count',
'card_ecommerce_to_noncommerce_ratio', 'card_mcc_diversity_count',
'ABM_cash_weight', 'transaction_count_EMT', 'transaction_count_EFT',
'birth_business_age', 'card_top_mcc_weight', 'account_age'
Moderate (0.35-0.65): 'debit_transaction_amount'

Main Province: Mostly Ontario and Other

Cluster 3:

High (0.65-1.0): 'account_age', 'birth_business_age', 'multi_country_transactions',
'card_top_mcc_weight', 'median_time_between_transactions',
'std_time_between_transactions', 'transaction_count_EFT', 'credit_share',
'Averagetime_between_transaction_day', 'transaction_count_EMT'
Moderate (0.35-0.65): 'ABM_cash_weight', 'transaction_count_ABM',
'pct_transactions_top10_busiest_days', 'unique_transaction_provinces'

Main Province: Other

Cluster 4:

High (0.65-1.0): 'multi_country_transactions', 'transaction_count_EMT',
'card_top_mcc_weight', 'transaction_count_EFT', 'birth_business_age'

Moderate (0.35-0.65): 'median_time_between_transactions', 'unique_transaction_provinces', 'ABM_cash_weight', 'std_time_between_transactions', 'credit_share', 'transaction_count_ABM', 'Averagetime_between_transaction_day'

Main Province: Manitoba

Cluster 5:

High (0.65-1.0): 'multi_country_transactions', 'transaction_count_EMT', 'card_top_mcc_weight', 'transaction_count_EFT', 'birth_business_age', 'ABM_cash_weight', 'unique_transaction_provinces'

Moderate (0.35-0.65): 'account_age', 'median_time_between_transactions', 'transaction_count_ABM', 'credit_share', 'std_time_between_transactions', 'Averagetime_between_transaction_day', 'card_mcc_diversity_count'

Main Province: Alberta

Cluster 6:

High (0.65-1.0): 'transaction_count_EMT', 'multi_country_transactions', 'card_top_mcc_weight', 'transaction_count_EFT', 'birth_business_age'

Moderate (0.35-0.65): 'median_time_between_transactions', 'unique_transaction_provinces', 'ABM_cash_weight', 'transaction_count_ABM', 'std_time_between_transactions', 'credit_share', 'account_age', 'Averagetime_between_transaction_day'

Main Province: New Brunswick

Cluster 7:

High (0.65-1.0): 'transaction_count_EFT', 'multi_country_transactions', 'card_top_mcc_weight', 'birth_business_age', 'transaction_count_EMT', 'credit_share', 'median_time_between_transactions']

Moderate (0.35-0.65): 'std_time_between_transactions', 'Averagetime_between_transaction_day', 'transaction_count_Cheque', 'unique_transaction_provinces', 'pct_transactions_top10_busiest_days'

Main Province: Saskatchewan

Recluster(2nd round clustering, k=7)

business_0:

High (0.65-1.0): 'Averagetime_between_transaction_day', 'std_time_between_transactions', 'pct_transactions_top10_busiest_days', 'credit_share', 'median_time_between_transactions', 'account_age'

Main Province: British Columbia

business_1:

High (0.65-1.0): 'median_time_between_transactions', 'Averagetime_between_transaction_day', 'std_time_between_transactions', 'credit_share'

Moderate (0.35-0.65): 'account_age'

Main Province: Ontario

business_2:

High (0.65-1.0): 'birth_business_age', 'transaction_count_EMT', 'credit_share', 'Averagetime_between_transaction_day', 'median_time_between_transactions'

Moderate (0.35-0.65): 'std_time_between_transactions'

Main Province: Seperated across multiple provinces

business_3:

High (0.65-1.0): 'transaction_count_ABM', 'transaction_count_Card', 'unique_transaction_cities', 'unique_transaction_provinces', 'multi_country_transactions', 'card_ecommerce_count', 'card_non_ecommerce_count', 'card_ecommerce_to_noncommerce_ratio', 'card_mcc_diversity_count', 'card_top_mcc_weight'

Moderate (0.35-0.65): 'account_age', 'median_time_between_transactions'

Main Province: Seperated across multiple provinces

business_4:

High (0.65-1.0): 'credit_share', 'Averagetime_between_transaction_day', 'median_time_between_transactions', 'std_time_between_transactions'

Moderate (0.35-0.65): "account_age"

Main Province: Alberta

business_5:

High (0.65-1.0): 'total_transaction_count', 'transaction_count_Cheque',
'transaction_count_EFT', 'debit_transaction_amount', 'credit_transaction_amount',
'average_debit_transaction_amount', 'average_credit_transaction_amount', 'credit_share',
'account_age'

Moderate (0.35-0.65): 'pct_transactions_top10_busiest_days'

Main Province: Separated across multiple provinces

business_6:

High (0.65-1.0): 'account_age', 'median_time_between_transactions',
'Averagetime_between_transaction_day', 'std_time_between_transactions', 'credit_share'

Moderate (0.35-0.65): 'pct_transactions_top10_busiest_days'

Main Province: Separated across multiple provinces